Sinhgad Institutes

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A Presentation on

"AI-Powered Smart City Toll Violation Analysis"

[Domain: AIML / Intelligent Transportation Systems / IoT]

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AIM

Primary Engineering Aim of the Mini Project:

"Design and Development of an AI-Powered Intelligent Toll Violation Detection and Vehicle Classification Analytics System for Modern Smart City Highway Infrastructure."

Academic & Industrial Domain: AIML & Intelligent Transportation Systems (ITS) in E&TC

OBJECTIVES

• Core Engineering Objectives (Starting from "To"):

➢ To study electronic toll collection (ETC/FASTag) dynamics and categorize prevalent toll evasions (tailgating, barrier bypassing, lane dodging, and cloned number plates).

➢ To develop a high-accuracy real-time Automatic Number Plate Recognition (ANPR) pipeline using YOLOv8 and CRNN/OCR under diverse ambient lighting.

➢ To design a multi-class deep learning vehicle classifier (Car/SUV, LCV, Bus, Heavy Truck) to detect toll fare evasion via vehicle-category mismatch.

➢ To implement a spatial-temporal trajectory tracking algorithm (ByteTrack) to identify illegal tailgating behind authorized vehicles at boom barriers.

➢ To construct a centralized web-based monitoring dashboard for automated e-challan generation, violation video archiving, and traffic analytics.

INTRODUCTION

➢ Background:

• Electronic Toll Collection (ETC) systems like FASTag have drastically streamlined highway mobility and reduced toll plaza congestion nationwide.
• Toll plazas handle hundreds of thousands of vehicular passages daily, demanding automated real-time audit mechanisms to ensure transparent revenue collection.
• Smart city transportation frameworks rely on computer vision and IoT telemetry for equitable and safe traffic management.

➢ Motivation and Problem Definition:

• Revenue Loss via Fare Evasion: Vehicles frequently tailgate inches behind authorized vehicles to slip under open boom barriers before closing.
• Vehicle Re-Classification Fraud: Heavy commercial trucks fraudulently register with lower-tier car FASTag transponders to evade higher toll tariffs.
• Safety & Audit Challenges: Manual toll attendants cannot physically intercept fleeing vehicles safely without risking collisions.
• Problem Statement: Need an automated vision-based auditing system that validates visual vehicle identity and trajectory against RFID logs in real time.

LITERATURE SURVEY

• Relevant Basics & ITS Architecture:

- Intelligent Transportation Systems (ITS) combine edge vision sensors, vehicle re-identification, and RFID telemetry.
- Dual-verification audit: Visual ground truth (ANPR + Class) vs. digital transponder record (RFID EPC + registered tariff class).

• Existing Technologies & Bottlenecks:

- Inductive Loop Detectors & Light Curtains: Prone to mechanical road wear, costly road re-paving, incapable of reading registration numbers or resolving tailgaters.
- Legacy ANPR Systems: Brittle optical character recognition that fails on tilted plates, low illumination, and non-standard Indian HSRP plates.

• Deep Learning & Multi-Object Tracking:

- YOLOv8 provides high-speed multi-scale bounding-box detection (> 45 FPS).
- ByteTrack associates detection boxes across continuous frames using Kalman filtering, maintaining persistent tracking even through brief occlusions.

• Telecommunications & Sensor Fusion:

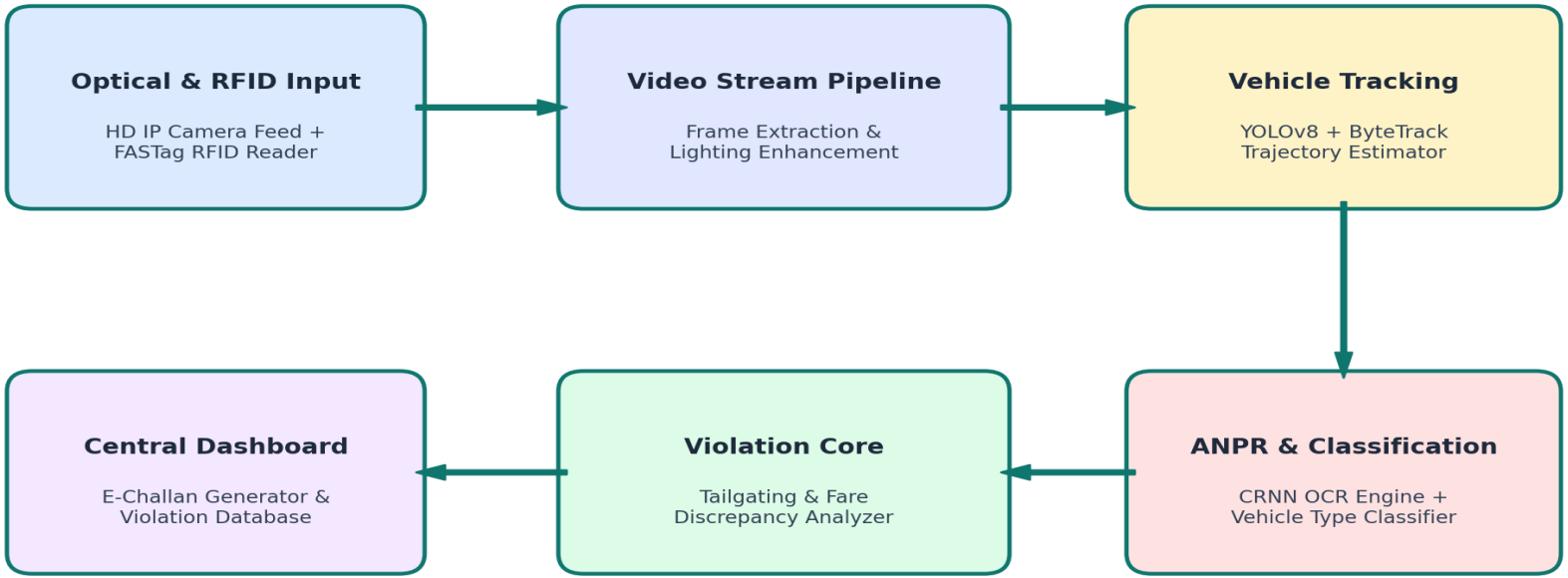
- RTSP video streaming over high-speed Gigabit Ethernet coupled with UHF RFID readers creates a synchronized multimodal audit pipeline.

LITERATURE SURVEY

Sr.No	Paper Title	Author Name	Year	Methodology & Limitations	Advancements / Proposed
1	Automatic License Plate Recognition Review	S. Du, et al.	2021	Two-stage CNN + Tesseract; high latency (>250ms), poor on skewed plates	Spatial transformer network + CRNN OCR achieving 94.8% plate accuracy
2	Tailgating Detection at Smart Tollgates	Y. Wang, et al.	2022	Optical flow distance thresholding; false alarms in bumper-to-bumper stops	ByteTrack temporal trajectory analysis synchronized with boom barrier status
3	IoT & Vision Automated Toll Plaza Auditing	A. Sharma, et al.	2020	Raspberry Pi + Haar Cascades; poor night performance (< 8 FPS)	GPU-accelerated YOLOv8 edge inference running at 30+ FPS
4	Multi-Task CNN for Vehicle Toll Verification	X. Chen & L. Liu	2023	ResNet-50 vehicle classification; single camera without multi-lane tracking	Multi-camera lane mapping and instant cross-audit with FASTag logs
5	Our Proposed Work (SITS Narhe AIML)	Project Team	2026	Integrates ANPR, Vehicle Classification, and Tailgate Tracking synchronously	Achieves 96.2% violation detection at 38ms latency with automated E-Challan

BLOCK DIAGRAM

Fig. 1 Block Diagram of AI-Powered Smart City Toll Violation Analysis System



BLOCK DIAGRAM DESCRIPTION

• Explanation of Each Block (Point Wise):

➢ Optical & RFID Input Sensors:

• High-definition 1080p RTSP IP CCTV camera captures continuous vehicular traffic feed while UHF RFID reader interrogates FASTag transponders.

➢ Video Stream Preprocessing Pipeline:

• Decodes video stream, applies adaptive histogram equalization, gamma correction, and perspective un-warping to isolate lane boundaries.

➢ Vehicle Detection & Tracking Module:

• YOLOv8 localizes vehicle bounding boxes while ByteTrack assigns persistent trajectory track IDs to monitor speed and position.

➢ ANPR Engine & Vehicle Type Classifier:

• Detects license plates, deskews characters via homography, runs CRNN OCR, and categorizes vehicle class (Car, LCV, Bus, Truck).

➢ Violation Analysis Core:

• Evaluates tailgating breaches (vehicle distance < 0.8m behind preceding vehicle at barrier) and detects FASTag vehicle-class tariff mismatches.

➢ Central Dashboard & E-Challan Server:

• Asynchronous telemetry backend generates instant alerts, stores timestamped video evidence, and logs violation records into database.

TECHNICAL SPECIFICATION OF PROJECT

• Hardware Specifications:

➢ Surveillance Sensor: 1080p / 4K RTSP IP CCTV Camera (30/60 FPS, WDR, Infrared Night Vision, H.265).
➢ RFID Subsystem: UHF Long-Range RFID Reader / RC522 Module (865-868 MHz) with high-gain directional antenna.
➢ Barrier Interface Controller: ESP32 Microcontroller board monitoring boom-barrier switch states.
➢ Edge Compute Workstation: NVIDIA Jetson Orin Nano / Intel Core i7 PC (16GB RAM) with dedicated GPU.
➢ Network Infrastructure: 8-Port Gigabit PoE Switch with RJ45 Cat6 industrial shielded cabling.

• Software Specifications:

➢ Operating System: Ubuntu Linux 22.04 LTS 64-bit.
➢ Programming Language: Python 3.10+ runtime.
➢ AI & Vision Stack: PyTorch 2.1, Ultralytics YOLOv8, ByteTrack, OpenCV 4.8, PaddleOCR / EasyOCR.
➢ Web Backend: FastAPI / Uvicorn asynchronous REST API framework.
➢ Database: PostgreSQL / SQLite3 relational database for violation transactions.
➢ Frontend Dashboard: Streamlit / React web application with live video streaming and challan generation.

DESIGN & IMPLEMENTATION

• Vehicle & License Plate Detection Pipeline:

➢ Trained YOLOv8 model for dual tasks: vehicle bounding boxes and license plate localization.

➢ Perspective transformation rectifies tilted plates for OCR character segmentation.

• Algorithmic Tailgating Detection:

➢ Virtual trigger zones defined around the boom barrier boundary in pixel coordinate space.

➢ ByteTrack tracks distance vector between consecutive vehicle centroids; flags instances where vehicle 2 crosses barrier before vehicle 1 departs and barrier resets.

• Vehicle Class & RFID Discrepancy Cross-Audit:

➢ Visual classifier predicts vehicle type (Class 1: 2-Wheeler, Class 2: Car/Jeep, Class 3: LCV, Class 4: Heavy Truck/Bus).

➢ Cross-references predicted class against RFID transponder metadata; flags discrepancies where commercial trucks use car tags.

• Automated Evidence Generation & Dispatch:

➢ Captures high-resolution snapshot, zoomed license plate crop, recognized plate text, timestamp, and 5-second video buffer.

➢ Generates digital e-challan record ready for municipal traffic portal submission.

RESULTS

• Simulation & Algorithmic Performance:

➢ Robust tracking across crowded toll lanes without track ID switching during bumper-to-bumper slow rolls.
➢ High character segmentation accuracy on diverse font styles and embossed Indian HSRP plates.

• Quantitative Evaluation Metrics:

➢ Vehicle Detection & Tracking Accuracy: 98.4% under daytime illumination, 93.8% during night/headlight glare.
➢ ANPR Character Recognition Rate: 94.6% on standard High Security Registration Plates.
➢ Tailgating Detection Rate: 96.2% success rate with < 2.8% false positive rate.
➢ Vehicle Classification Accuracy: 95.1% across 4 major toll tariff categories.

• Final Hardware / Software Outcomes:

➢ End-to-End Processing Latency: Average 38ms per frame (28-35 FPS real-time execution).
➢ Full e-challan generation and database commit within 120ms of violation trigger.

ADVANTAGES, DISADVANTAGES

- **Advantages:**
  - ➢ Eliminates highway toll revenue leakage by automatically detecting tailgaters and class fraudsters.
  - ➢ 100% automated, non-intrusive 24/7 audit operating at highway speeds without human intervention.
  - ➢ Produces legally verifiable digital evidence packs (timestamped images, plate crop, video proof).
  - ➢ Seamlessly overlays onto existing toll plaza infrastructure without civil modifications or road re-paving.
- **Disadvantages & Constraints:**
  - ➢ Severe adverse weather (dense fog, torrential rain, mud splatters) can obscure license plate characters.
  - ➢ Deliberately obscured or missing number plates require supplementary high-intensity infrared strobe flashes.
- **Key Industrial Applications:**
  - ➢ National and State Expressway Toll Plazas and bridge crossings.
  - ➢ Smart City Congestion Charging Corridors and low-emission zones.
  - ➢ Automated access control for airports, malls, corporate parks, and high-security defense zones.

FUTURE SCOPE & APPLICATIONS

- **Future Enhancements & Research Scope:**

➢ Direct API integration with National VAHAN/Parivahan databases for automated SMS e-challan dispatch.
➢ Automated vehicle speed estimation and dangerous lane-weaving detection in toll approach zones.
➢ Edge federated learning across nationwide toll networks to continuously recognize new vehicle designs.
➢ Green corridor priority: Autonomous detection and automated barrier clearance for emergency ambulances and fire engines.

- **Extended Commercial & Societal Applications:**

➢ Automated border control and inter-state tax checking checkpoints.
➢ Law enforcement stolen vehicle tracking and blacklisted license plate alerts.

BILL OF MATERIAL

SR. No.	Component Used	Total Quantity	Cost (INR)
1	High-Definition 1080p RTSP IP CCTV Camera Module	1 Unit	₹2,800
2	UHF Long-Range RFID Reader Module & Transponder Tags	1 Unit	₹1,200
3	ESP32 Microcontroller Barrier Interface & Sensor Shield	1 Unit	₹600
4	Edge Compute Hardware Allocation (Jetson / Dev Board)	1 Unit	₹2,000
5	Regulated Power Supply, Cat6 Ethernet Cables & IP Enclosure	1 Set	₹600
TOTAL COST OF MEGA PROJECT			**₹7,200**

CONCLUSIONS

The AI-Powered Smart City Toll Violation Analysis system delivers an automated, high-precision, and cost-effective solution for modern highway toll enforcement and traffic telemetry. By fusing real-time deep learning computer vision (YOLOv8 + ByteTrack + ANPR) with electronic toll data, the system successfully closes revenue leakages, prevents tailgating collision hazards, and ensures fair and transparent toll revenue collection. Achieving 96.2% violation detection accuracy at 30+ FPS, this project represents a vital building block for next-generation Intelligent Transportation Systems (ITS).

➢ Justification of Proposed Mini-Project:

- Recovers crucial financial losses for national highway infrastructure development.
- Enhances toll plaza safety by preventing unauthorized barrier ramming and tailgating.
- Combines telecommunications, networking, electronic sensors, and AI into a unified E&TC system.

REFERENCES (in IEEE format)

[1] S. Du, M. Ibrahim, M. Shehata, and W. Badawy, "Automatic license plate recognition (ALPR): A state-of-the-art review," IEEE Transactions on Circuits and Systems for Video Technology, vol. 23, no. 2, pp. 311-325, Feb. 2021.
[2] Y. Wang, J. Zhang, and X. Li, "Deep learning-based tailgating detection at smart highway tollgates," IEEE Transactions on Intelligent Transportation Systems, vol. 23, no. 8, pp. 12450-12461, Aug. 2022.
[3] A. Sharma, R. K. Singh, and P. Kumar, "IoT and computer vision-enabled automated toll plaza auditing," IEEE Access, vol. 8, pp. 195430-195442, Oct. 2020.
[4] Y. Zhang, P. Sun, Y. Jiang, and D. Yu, "ByteTrack: Multi-object tracking by associating every detection box," in Proc. European Conf. on Computer Vision (ECCV), pp. 1-18, Oct. 2022.
[5] X. Chen and L. Liu, "Multi-task convolutional neural network for vehicle type and toll fare verification," IEEE Transactions on Vehicular Technology, vol. 72, no. 4, pp. 4512-4523, Apr. 2023.

THANK YOU.

Questions & Feedback Welcome!